

KHILOK, Russian Federation

WMO: 308440

Lat: 51.367N

Lon: 110.467E

Elev: 802

StdP: 92.06

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.8	(c) -36.4	(d) -41.0	(e) 0.1	(f) -38.5	(g) -38.6	(h) 0.1	(i) -36.2	(j) 12.4	(k) -16.4	(l) 11.2	(m) -15.8	(n) 0.4	(o) 90	(p) 0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	29.8	17.4	27.6	16.6	25.6	15.9	19.3	26.2	18.2	24.5	17.3	23.0	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.6	21.5	16.1	12.6	20.4	15.2	11.9	19.6	58.8	26.0	55.2	24.7	52.0	22.7	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.6	(b) 9.3	(c) 7.9	(e) -42.4	(f) 33.5	(g) 1.8	(h) 2.3	(i) -43.6	(j) 35.2	(k) -44.7	(l) 36.6	(m) -45.7	(n) 37.9	(o) -46.9	(p) 39.6		
(a) 10.6	(b) 9.3	(c) 7.9	(e) -42.2	(f) 21.2	(g) 2.0	(h) 1.2	(i) -43.7	(j) 22.1	(k) -44.9	(l) 22.8	(m) -46.0	(n) 23.5	(o) -47.5	(p) 24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.8	-23.5	-19.0	-9.2	0.9	7.6	14.6	17.8	14.8	7.9	-1.1	-12.5	-20.9
	DBStd	15.07	5.79	5.87	6.34	4.71	4.01	3.68	3.00	3.41	4.01	4.85	6.27	6.43
	HDD10.0	4892	1038	812	595	274	97	6	0	4	87	345	675	959
	HDD18.3	7389	1297	1045	853	523	333	119	46	116	312	603	925	1217
	CDD10.0	588	0	0	0	1	22	145	241	154	25	0	0	0
	CDD18.3	43	0	0	0	0	0	8	28	7	0	0	0	0
	CDH23.3	968	0	0	0	1	32	312	448	156	20	0	0	0
	CDH26.7	272	0	0	0	0	5	89	141	34	3	0	0	0
Wind	WSAvg	2.8	2.4	2.5	2.9	3.6	3.6	2.8	2.5	2.4	2.8	2.8	2.8	2.3
Precipitation	PrecAvg	380	5	4	7	15	25	61	99	81	45	17	10	9
	PrecMax	555	13	12	23	38	63	112	203	185	103	49	24	36
	PrecMin	208	0	0	0	1	4	6	30	18	1	1	2	1
	PrecStd	90	3	3	6	10	15	32	42	38	27	12	6	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.4	0.9	11.1	20.2	27.3	32.8	33.0	30.1	26.3	17.2	5.1	-3.7
		MCWB	-6.6	-3.0	3.6	8.5	13.2	17.2	19.0	17.7	14.4	8.1	0.7	-5.4
	2%	DB	-9.1	-3.0	7.0	16.1	24.0	29.3	30.3	27.0	22.1	12.9	1.8	-6.5
		MCWB	-10.3	-5.5	1.0	6.5	11.5	15.8	18.5	17.0	12.8	5.7	-1.4	-7.7
	5%	DB	-11.6	-5.6	4.1	13.0	21.1	27.1	28.0	24.9	19.2	9.8	-0.6	-8.4
		MCWB	-12.6	-7.6	-0.8	4.6	9.9	15.4	17.9	16.7	11.3	4.0	-2.8	-9.4
	10%	DB	-14.0	-8.1	1.4	10.1	18.2	24.8	25.8	22.6	16.3	6.7	-2.7	-10.7
		MCWB	-14.7	-9.6	-2.3	3.3	8.5	14.8	17.1	15.8	10.0	2.3	-4.4	-11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.6	-2.9	3.9	8.9	14.0	19.5	21.2	19.7	15.3	8.4	0.9	-5.4
		MCDB	-5.4	0.4	10.5	18.8	25.6	26.0	28.3	26.1	22.9	16.4	4.9	-4.0
	2%	WB	-10.2	-5.4	1.4	6.9	12.1	17.8	19.9	18.3	13.7	6.0	-1.0	-7.6
		MCDB	-9.2	-3.2	6.4	15.0	22.1	25.7	27.3	24.3	20.0	12.0	1.4	-6.5
	5%	WB	-12.6	-7.6	-0.5	5.1	10.7	16.5	18.9	17.3	12.3	4.2	-2.7	-9.4
		MCDB	-11.7	-5.7	3.8	12.2	19.7	23.8	25.7	22.8	17.3	8.9	-0.8	-8.4
	10%	WB	-14.7	-9.6	-2.1	3.5	9.2	15.5	18.0	16.4	10.7	2.7	-4.3	-11.5
		MCDB	-14.0	-8.1	1.3	9.4	16.8	22.4	23.7	21.4	15.6	6.4	-2.7	-10.7
Mean Daily Temperature Range	5% DB	MDBR	13.3	17.3	16.4	13.5	15.5	15.4	13.6	13.2	14.0	12.7	12.3	11.7
		MCDBR	13.2	19.5	18.9	19.9	23.2	22.1	19.2	18.2	20.5	18.4	13.6	10.6
		MCWBR	12.3	16.9	13.1	11.1	11.8	10.0	8.6	9.1	11.1	11.3	10.7	9.7
	5% WB	MCDBR	13.0	18.9	18.0	18.1	20.4	17.2	15.1	15.0	17.3	17.0	12.9	10.7
		MCWBR	12.1	16.5	12.5	10.3	10.7	8.3	7.4	8.0	9.7	10.6	10.3	9.8
Clear-Sky Solar Irradiance	taub	0.259	0.283	0.317	0.371	0.459	0.449	0.410	0.374	0.335	0.301	0.281	0.263	
	taud	2.190	2.172	2.199	2.218	1.987	2.130	2.310	2.394	2.448	2.459	2.302	2.169	
	Ebn at Noon	753	837	875	864	797	810	839	855	855	822	725	674	
	Edh at Noon	70	95	114	127	168	147	121	106	89	72	63	61	
All-Sky Solar Radiation	RadAvg	1.48	2.61	4.13	5.13	5.71	5.97	5.54	4.73	3.69	2.41	1.43	1.10	
	RadStd	0.06	0.08	0.15	0.40	0.35	0.39	0.46	0.30	0.30	0.18	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page